Roll No. | | | | | | | |

End Term Examination-2025

Name of the Course: B. Tech. Biotechnology/B. Tech. Biotechnology (Hons)
Semester: IV
Name of the Paper: Recombinant DNA Technology
Time: 3 Hours

Paper Code: TBT401
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty. Each question carries 10 marks.

Q1	(20marks)	**CO 1**
(a)	Describe design of biotechnology laboratory suitable for BSL-2 or BSL-3 research. Justify your design choices based on biosafety principles, including containment, workflow, and decontamination procedures	
(b)	Discuss the scope and applications of recombinant DNA technology (RDT) in improving human health and quality of life.	
(c)	Analyze the steps involved in gene cloning. What are the roles of vectors, enzymes, and host cells in this process?	
Q2	(20 marks)	**CO 2**
(a)	Write short notes on any two of the followings: a)- Modifying enzymes in RDT; b)- λ vector; c)- α complementation	
(b)	Draw a labelled diagram of pUC18 plasmid vector and describe its cloning and screening strategies.	
(c)	How would you use the pJB8 (Cosmid) vector to clone a large genomic fragment? Outline the steps for both cloning and screening of recombinants	
Q3	(20 marks)	**CO 3**
(a)	Describe principle and process of Red-White screening in YAC vectors.	
(b)	Explain the role of a promoter in gene expression. Describe the key components of an expression vector and their functions."	
(c)	How would you use the regulatory features of the pET-28b vector to control expression of a recombinant protein in E. coli? Outline the steps involved in induction and expression.	
Q4	(20 marks)	**CO 4**
(a)	How would you apply Sanger sequencing to identify the sequence of any unknown DNA? Describe the steps involved and the expected output.	
(b)	Compare and contrast the principles, detection chemistries, and applications of conventional PCR and Real-Time PCR. What are the strengths and limitations of each?	
(c)	How would you introduce a point mutation in a plasmid using site-directed mutagenesis? Outline the steps involved and explain the purpose of each.	
Q5	(20 marks)	**CO 5**
(a)	How would you apply the DNA fingerprinting technique in a forensic investigation? Describe the steps involved and how the results can be used for identification.	
(b)	How would you apply recombinant DNA technology to produce insulin in a laboratory setting? Describe the steps and the role of host organisms, vectors, and expression systems.	
(c)	Discuss the principle of CRISPR-CAS9 gene editing technology along with its application in gene therapy.	